Task 3: Data import and export operations between HDFS and local file system.

Date of the Session: ----/-----/------ Time of the Session: to

Pre Lab:

1. What is the purpose of data import and export operations between the local file system and HDFS?

2. Explain the command hdfs dfs -put and mention one alternative command for the same operation.

3. What happens internally when a file is copied from the local file system to HDFS?

4. What is the purpose of the hdfs dfs -cat command, and when is it useful?

5. Differentiate between hdfs dfs -copyToLocal and hdfs dfs -get.

In-Lab:

Import and export operations:

a.Writing a file from local file system to Hadoop distributed file system

hdfs dfs –put /home/cloudera/sobhana/friends.txt /user/cloudera/mydir/

```
[cloudera@quickstart ~]$ hdfs dfs -put /home/cloudera/navya/friends.txt /user/cloudera/mydir/
put: `/user/cloudera/mydir/friends.txt': File exists
```

b. Writing a file from local filesystem to Hadoop distributed file system (using copy From Local)

- Create a file weather.txt in local file system

```
[cloudera@quickstart navya]$ gedit weather.txt
[cloudera@quickstart navya]$ cat weather.txt
Weather is the our to hor state of atmosphere at specific place and time.
YEAR    TEMPERATURE
2021      45
2022      46
2023      42
2024      40
2025      50
```

- Copy file from local filesystem to Hadoop distributed filesystem
- View the files in hdfs mydir directory

```
[cloudera@quickstart navya]$ hdfs dfs -copyFromLocal weather.txt /user/cloudera/mydir/
copyFromLocal: `/user/cloudera/mydir/weather.txt': File exists
[cloudera@quickstart navya]$ hdfs dfs -ls /user/cloudera
Found 3 items
drwxr-xr-x   - cloudera cloudera          0 2026-01-26 22:05 /user/cloudera/mydir
drwxr-xr-x   - cloudera cloudera          0 2026-01-26 21:51 /user/cloudera/navya
drwxr-xr-x   - cloudera cloudera          0 2026-01-19 23:02 /user/cloudera/task2
```

c.View the file contents.

–cat /user/cloudera/mydir/friends.tx

```
[cloudera@quickstart ~]$ hdfs dfs -cat /user/cloudera/mydir/friends.txt
NavyaSri Sarvani
```

d. Create an empty file in HDFS and list the files

hdfs dfs –touchz/user/cloudera/mydir/file1.txt

```
[cloudera@quickstart ~]$ hdfs dfs -touchz /user/cloudera/mydir/file1.txt
[cloudera@quickstart ~]$ hdfs dfs -ls /user/cloudera/mydir
Found 3 items
-rw-r--r--   1 cloudera cloudera          0 2026-01-26 22:17 /user/cloudera/mydir/file1.txt
-rw-r--r--   1 cloudera cloudera         17 2026-01-26 21:53 /user/cloudera/mydir/friends.txt
-rw-r--r--   1 cloudera cloudera        159 2026-01-26 22:05 /user/cloudera/mydir/weather.txt
```

e. Reading a file from HDFS to the local file system and list the files in local filesystem.

hdfs dfs –copyToLocal/user/cloudera/mydir/file1.txt/home/cloudera/sobhana/

```
[cloudera@quickstart navyasri]$ ls
file1.txt  friends.txt  weather.txt
```

Post-Lab:

1. Describe how the NameNode processes the command hdfs dfs -mkdir -p /user/cloudera/mydir. What types of data are generated, and how does HDFS handle parent directories if they are missing?

2. Why does the command hdfs dfs -ls / only communicate with the NameNode rather than with DataNodes?

3. What happens when the instruction hdfs dfs -ls -R / is executed? For systems with millions of files, how does this command affect the NameNode?

4. How is hdfs dfs -rm different from hdfs dfs -rm -r? How does the Trash feature affect deletion?

5. Does HDFS create physical or logical copies during hdfs dfs -cp source destination? Explain block copying and placement.

(For Evaluator's use only)

Comment of the Evaluator (if Any)	Evaluator's Observation Marks Secured:_____ out of _____ Signature of the Evaluator